

We now use the quotient rule, differentiating implicitly as follows:

$$\begin{aligned} y'' &= \frac{(4y^3 + 3) D_x(12x^2 + 5) - (12x^2 + 5) D_x(4y^3 + 3)}{(4y^3 + 3)^2} \\ &= \frac{(4y^3 + 3)(24x) - (12x^2 + 5)(12y^2 y')}{(4y^3 + 3)^2} \end{aligned}$$

Substituting for y' yields

$$\begin{aligned} y'' &= \frac{(4y^3 + 3)(24x) - (12x^2 + 5) \cdot 12y^2 \left(\frac{12x^2 + 5}{4y^3 + 3} \right)}{(4y^3 + 3)^2} \\ &= \frac{(4y^3 + 3)^2(24x) - 12y^2(12x^2 + 5)^2}{(4y^3 + 3)^3}. \end{aligned}$$

EXERCISES 3.7

Exer. 1–18: Assuming that the equation determines a differentiable function f such that $y = f(x)$, find y' .

1 $8x^2 + y^2 = 10$

2 $4x^3 - 2y^3 = x$

3 $2x^3 + x^2y + y^3 = 1$

4 $5x^2 + 2x^2y + y^2 = 8$

5 $5x^2 - xy - 4y^2 = 0$

6 $x^4 + 4x^2y^2 - 3xy^3 + 2x = 0$

7 $\sqrt{x} + \sqrt{y} = 100$

8 $x^{2/3} + y^{2/3} = 4$

9 $x^2 + \sqrt{xy} = 7$

10 $2x - \sqrt{xy} + y^3 = 16$

11 $\sin^2 3y = x + y - 1$

12 $x = \sin(xy)$

13 $y = \csc(xy)$

14 $y^2 + 1 = x^2 \sec y$

15 $y^2 = x \cos y$

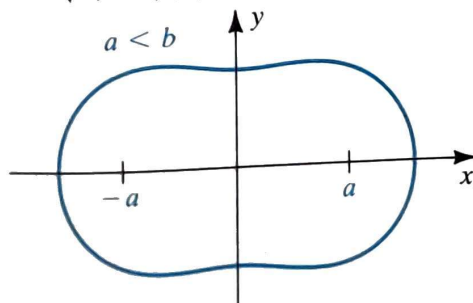
16 $xy = \tan y$

17 $x^2 + \sqrt{\sin y} - y^2 = 1$

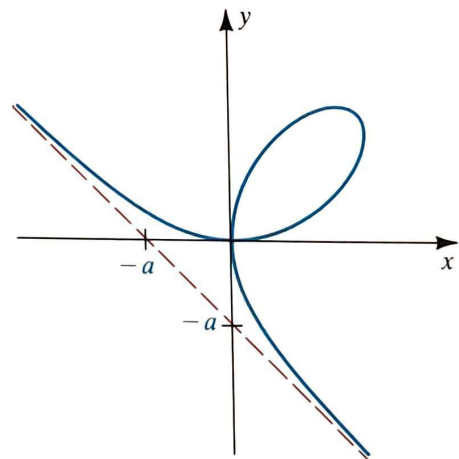
18 $\sin \sqrt{y} - 3x = 2$

Exer. 19–22: The equation of a classical curve and its graph are given for positive constants a and b . (Consult books on analytic geometry for further information.) Find the slope of the tangent line at the point P for the stated values of a and b .

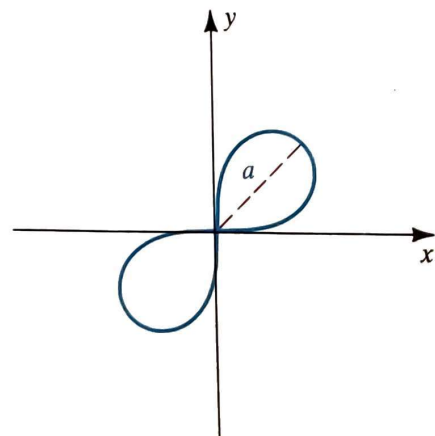
19 *Ovals of Cassini*: $(x^2 + y^2 + a^2)^2 - 4a^2x^2 = b^4$;
 $a = 2$, $b = \sqrt{6}$, $P(2, \sqrt{2})$



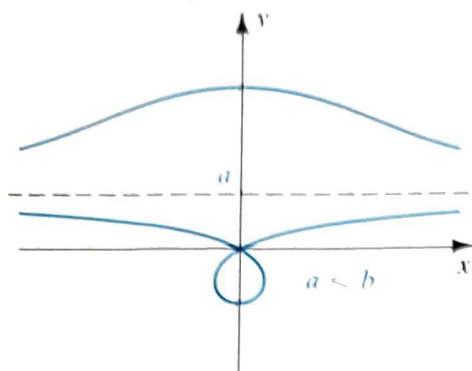
20 *Folium of Descartes*: $x^3 + y^3 - 3axy = 0$;
 $a = 4$, $P(6, 6)$



21 *Lemniscate of Bernoulli*: $(x^2 + y^2)^2 = 2a^2xy$;
 $a = \sqrt{2}$, $P(1, 1)$



- 22 Conchoid of Nicomedes: $(y - a)^2(x^2 + y^2) = b^2 y^2$;
 $a = 2$, $b = 4$, $P(\sqrt{15}, 1)$



Exer. 23–28: Find the slope of the tangent line to the graph of the equation at P .

- 23 $xy + 16 = 0$; $P(-2, 8)$
 24 $y^2 - 4x^2 = 5$; $P(-1, 3)$
 25 $2x^3 - x^2y + y^3 - 1 = 0$; $P(2, -3)$
 26 $3y^4 + 4x - x^2 \sin y - 4 = 0$; $P(1, 0)$
 27 $x^2y + \sin y = 2\pi$; $P(1, 2\pi)$
 28 $xy^2 + 3y = 27$; $P(2, 3)$

Exer. 29–34: Assuming that the equation determines a function f such that $y = f(x)$, find y'' , if it exists.

- 29 $3x^2 + 4y^2 = 4$ 30 $5x^2 - 2y^2 = 4$
 31 $x^3 - y^3 = 1$ 32 $x^2y^3 = 1$
 33 $\sin y + y = x$ 34 $\cos y = x$

Exer. 35–38: How many implicit functions are determined by the equation?

- 35 $x^4 + y^4 - 1 = 0$ 36 $x^4 + y^4 = 0$
 37 $x^2 + y^2 + 1 = 0$ 38 $\cos x + \sin y = 3$

- 39 Show that the equation $y^2 = x$ determines an infinite number of implicit functions.
 40 Use implicit differentiation to show that if P is any point on the circle $x^2 + y^2 = a^2$, then the tangent line at P is perpendicular to OP .
 41 Suppose that $3x^2 - x^2y^3 + 4y = 12$ determines a differentiable function f such that $y = f(x)$. If $f(2) = 0$, use differentials to approximate the change in $f(x)$ if x changes from 2 to 1.97.
 42 Suppose that $x^3 + xy + y^4 = 19$ determines a differentiable function f such that $y = f(x)$. If $P(1, 2)$ is a point on the graph of f , use differentials to approximate the y -coordinate b of the point $Q(1.1, b)$ on the graph.
 [c] 43 Suppose that $x^2 + xy^3 = 4.0764$ determines a differentiable function f such that $y = f(x)$.
 [a] If $P(1.2, 1.3)$ and $Q(1.23, b)$ are on the graph of f , use (3.31) to approximate b .
 [b] Apply the method in (a), using $Q(1.23, b)$ to approximate the y -coordinate of $R(1.26, c)$. (This process, called *Euler's method*, can be repeated to approximate additional points on the graph.)
 [c] 44 Suppose that $\sin x + y \cos y = -2.395$ determines a differentiable function f such that $y = f(x)$.
 [a] If $P(2.1, 3.3)$ is an approximation to a point on the graph of f , use (3.31) to approximate the y -coordinate of $Q(2.12, b)$.
 [b] Apply the method in (a), using $Q(2.12, b)$ to approximate the y -coordinate of $R(2.14, c)$.

3.8 RELATED RATES

Suppose that two variables x and y are functions of another variable t , say

$$x = f(t) \quad \text{and} \quad y = g(t).$$

By (3.7)(ii), we may interpret the derivatives dx/dt and dy/dt as the rates of change of x and y with respect to t . As a special case, if f and g are position functions for points moving on coordinate lines, then dx/dt and dy/dt are the velocities of these points (see (3.2)). In other situations these derivatives may represent rates of change of physical quantities.

In certain applications x and y may be related by means of an equation, such as

$$x^2 - y^3 - 2x + 7y^2 - 2 = 0.$$